

DESIGN SUMMARY

Parameter	Value	Parameter	Value
Model number	VM-MD-B-305/500-AR	Material	Clinker
Capacity (req.)	100 t/h	Capacity (ach.)	279.3 t/h
Lift height	25 m	Belt / chain speed	1.83 m/s
Head pulley Ø	500 mm	Drive type	Belt
Bucket series	AC_12x8	Bucket W×P×Depth	305×203×216 mm
Motor power	37 kW	Head shaft Ø	108 mm
Casing width	356 mm	No. of buckets	225

FABRICATION BILL OF MATERIALS

All items identified by description (part numbers to be assigned at fabrication release). Mass values are calculated values — weigh actual parts at QC gate.

#	Description	Qty	Unit	Material / Grade	Mass ea. (kg)	Total (kg)	Weld / Fab spec
SH AF T							
1	Head shaft (solid section)	1	EA	ASTM A36 Mild Steel	62.1	62.1	Full pen butt weld
2	Boot shaft (solid section, free-running)	1	EA	ASTM A36 Mild Steel	52.6	52.6	Full pen butt weld
3	Head shaft key	1	EA	AISI 1018 (standard key stock)	0.0	0.0	Full pen butt weld
PU LL EY							
4	Head pulley shell	1	EA	S355JR	53.9	53.9	—
5	Head pulley end disc (annular plate)	2	EA	S355JR	35.3	70.6	—
6	Head pulley lagging — ceramic embedded rubber	1	EA	NR/SBR rubber	12.2	12.2	—
7	Boot / tail pulley assembly (shell + discs)	1	EA	S355JR	21.9	21.9	—
BE LT							
8	Elevator belt — 2 PLY	1	SET	EP 2 PLY rubber conveyor belt	422.1	422.1	—
9	Buckets — Series AC_12x8 305×216mm	225	EA	HDPE	11.0	2475.0	—
10	Bucket attachment bolt set — M12 stainless	225	SET	A2-70 stainless steel	0.1	19.1	—
DR IV E							
11	Drive motor — 37 kW IE3	1	EA	IE3 efficiency class	229.4	229.4	—
12	Helical-bevel gearbox	1	EA	Cast iron housing / alloy steel gears	388.5	388.5	—

#	Description	Qty	Unit	Material / Grade	Mass ea. (kg)	Total (kg)	Weld / Fab spec
TAKE-UP							
	13 Gravity take-up counterweight (ballast + frame)	1	SET	Cast iron / concrete ballast; S275 frame	3597.0	3597.0	—
CASING							
	14 Casing plate (fabricated panels)	1	SET	S275JR	2175.7	2175.7	Fillet weld 8mm leg
FASTENERS							
	15 Casing assembly bolt set — M12	793	SET	Zinc-plated steel, Class 8.8	0.1	43.6	Fillet weld 8mm leg
BEARINGS							
	16 Head shaft pillow block bearing assembly	2	EA	Chrome steel (GCr15)	6.8	13.5	Full pen butt weld
	17 Boot shaft pillow block bearing assembly	2	EA	Chrome steel (GCr15)	6.0	11.9	Full pen butt weld
CHUTE							
	18 Discharge chute fabrication (back, side plates)	1	SET	S275JR	22.8	22.8	—
	19 Chute wear liner — AR500 Wear Plate	1	SET	AR500 Wear Plate	33.5	33.5	—

QUALITY CONTROL CHECKPOINTS

#	Checkpoint	Method	Acceptance Criteria	Sign-off
1	Head shaft material cert	Review MTC	A36 / equiv. fy ≥ 250 MPa, Charpy 27J @ 0°C	
2	Head shaft dimensional check	Caliper / CMM	Ø108 +0/−0.02 mm, runout < 0.05 mm	
3	Keyway dimensions	Gauge	Width / depth per AS 1513, surface Ra ≤ 1.6 µm	
4	Head pulley shell weld	Dye penetrant / UT	No cracks, porosity ≤ 1.5 mm dia, full pen confirmed	
5	Casing panel flatness	Straight-edge	≤ 3 mm in any 1 m, mitre joints gapped < 1.5 mm	
6	Casing bolt pattern	Template check	Holes within ±0.5 mm of drawing, no burrs	
7	Belt joint / splice	Visual + tension test	Mechanical splice per CEMA, rated at 110 % working tension	
8	Bucket attachment torque	Torque wrench	M12 Gr.8.8 bolts at 85 Nm ± 5 %	
9	Alignment head-to-boot	Laser / plumb line	< 1 mm/m deviation over elevator height	
10	Paint / coating	DFT gauge	Min 75 µm DFT two-coat epoxy, no holidays on internal surfaces	

MATERIAL TEST CERTIFICATE REQUIREMENTS

Item	Std. / Grade	MTC Type	Min. Requirements
Head shaft	ASTM A36 / AS 3678-300	3.1 per EN 10204	Tensile, yield, elongation, Charpy impact @ 0°C
Casing plates	AS 3678-250 or 350	2.2 per EN 10204	Grade confirmation, thickness $\pm 10\%$
Bucket material	Carbon steel / AR400	3.1 per EN 10204 if AR	Hardness cert for AR grades, heat number traceability
Head pulley shell	AS 3678-350	2.2 per EN 10204	Grade, thickness, no laminar defects (UT if $t > 25\text{ mm}$)
Fasteners	ISO 898-1 Gr.8.8	2.1 per EN 10204	Grade marking on all fasteners, head date code visible